

Reflective Report

Gold and Oil Market Intelligence Project

Setting out

The brief framed this as more than a price analysis. I was asked to act as part of a data engineering and analytics team for a financial intelligence company, and to move past describing what gold and oil prices did toward explaining why they moved. I took that seriously from the start. What the project actually taught me is that the explaining only matters once you can trust what you are explaining, and that lesson did not arrive in a tidy way. It came from a problem I nearly walked past.

Starting with the pipeline

My first work was on the data pipeline, a layered Snowflake build that carried raw price files through a cleaning stage into an analytics layer, with hypothesis tests and an executive brief at the end. The structure was sound. It kept raw, clean and analytics schemas separate, logged each step to an audit table, and defined its own functions for the statistical tests. When I traced the dependencies carefully, though, I found that the view the whole analytics layer relied on, the market signals view, was never created anywhere in the scripts. Everything downstream read from it and nothing built it.

I rebuilt that view from the columns its consumers expected, and fixed three smaller breaks while I was there: a table defined twice with two different parsing methods, a reference to a table that never existed, and a stray line that would have stopped the script on the way through. None of this was glamorous. It was exactly the part of the work the brief had warned me about, that teams spend more time preparing and validating data than analysing it, and that building trust in the data is often worth more than building another dashboard.

The moment the project changed

The real turn came when I loaded the actual files. I had six datasets: daily gold and oil prices, monthly inflation, a daily dollar index, daily VIX, and annual world oil production. The obvious next step was to start joining them and testing relationships. Before that, I checked something basic that is easy to skip. Do these datasets even cover the same period?

They did not. The gold and oil price files run from December 2011 to December 2018. The dollar index and the VIX run from June 2021 to June 2026. They do not share a single month. The six market events I was meant to study were dated 2022 to 2024, after the price data ends.

This was the most important finding in the whole project, and it is one that produces no error. If I had run the pipeline as built, it would have returned empty correlation tables and a market snapshot that looked like real output but rested on joins of nothing. A reader would not have known. I only caught it because I plotted the coverage of each series on one timeline and looked at it.

Four of the six questions I had set out to answer, everything involving the dollar or market volatility, could not be answered with this data at all. Writing that down plainly, rather than presenting a confident but hollow result, is the single thing I am most glad I did.

What the data could actually show

What remained was still worth real analysis. Over the 2011 to 2018 window, 85 months in total, gold fell about 23 percent and oil about 51 percent. This was a commodity bear market, not the safe haven story people often expect from gold. Oil was about twice as volatile as gold from one month to the next. The relationship between the two was the most interesting result. On price levels they looked strongly linked, with a correlation near 0.64, but on monthly returns the correlation was almost zero, around 0.01, and the rolling twelve month version swung between strongly positive and strongly negative. The single headline number had been hiding the truth, and only a rolling view exposed it.

I could test two of the six hypotheses properly. Gold as an inflation hedge did not hold in this window. It actually did slightly worse in higher inflation months, and the difference was not statistically significant. The oil supply test could not be run as a fair comparison at all, because world output rose in all but one year of the period, which left a group of one and an undefined result. I reported both outcomes as they were, including the one that simply could not be computed.

I did not want to trust code on paper, so I ran the whole pipeline through and read its actual output tables rather than assume they would match my reasoning. They did. The quality counts, the hypothesis verdicts, and the empty tables where the data could not support a test all came out the same way I had found by hand. Seeing the pipeline produce its own empty results, instead of guessing they would be empty, was the proof I needed that the coverage problem was real and not a mistake in my own working.

Question	Answerable?	What the data showed
Did gold and oil move together?	Yes	Linked on price levels (0.64), almost not at all on monthly returns (0.01); the link shifts over time
Which was more volatile?	Yes	Oil, about twice gold, month to month
What happened to prices?	Yes	Gold fell about 23%, oil about 51%, across 2011 to 2018
Is gold an inflation hedge?	Yes	No measurable hedge in this window; the difference was not significant
Does oil rise when output falls?	No	Undefined: world output rose in all but one year, so no fair comparison
Does gold track the dollar?	No	No overlapping data between the price and dollar series
Is gold a safe haven under stress?	No	No overlapping VIX data to test it

Table 1. What the project set out to answer, and what the available data allowed.

The five levels, and thinking like an investigator

The course framework asked me to move through five levels: what happened, when, where, why, and what next. Mapping my work onto them was clarifying, and also honest about where it stopped. I could answer the what and the when from the price and volatility analysis. The why was only partly reachable, because the drivers theory points to first, the dollar and market stress, were precisely the ones I had no overlapping data for. The where level did not apply, since these are global benchmark prices with no geography, and I decided it was better to say so than to force a location onto data that has none. The what next turned into a set of recommendations whose first item is not another analysis but fixing the data coverage, so the open questions can finally be tested.

Working with AI honestly

The brief was direct about AI. It can help generate code and documentation, but it cannot replace critical thinking, and I should always ask whether the answer is correct and whether I have validated it. I used AI tooling through this project to draft and check code, and that stance turned out to be exactly right. The tools were quick at producing a pipeline and an analysis notebook, but the judgments that mattered were mine to make and to verify: noticing that a view was missing, deciding to plot coverage before trusting any join, and refusing to let an empty result stand as a finding. The validation was not a formality. It was where the real value sat.

What I would do differently

If I started again, I would run the coverage check first, before writing a single line of analysis, rather than partway through. I would be more careful with reused public datasets, since the gold and oil files turned out to cover a different era than the macro indicators, something I could have confirmed at the point of download. And I would treat the inflation hedge idea as conditional from the outset, rather than as something the data was likely to confirm.

Closing

The lasting lesson is simple to state and harder to practise. The most useful thing I produced was not a dashboard or a model. It was the decision to say clearly what the data could and could not support. The engineering was sound and the charts were drawn from real figures, but the project only became trustworthy once I was honest about its limits. That, more than any single number, is what I am taking from this work.